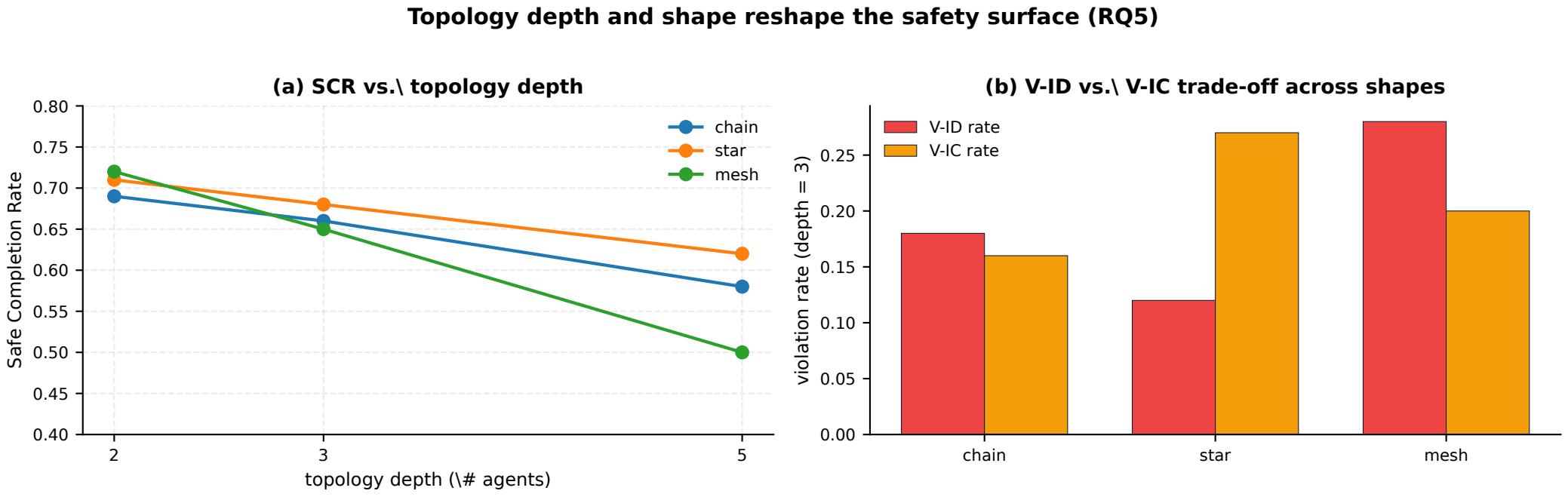
Topology depth and shape reshape the safety surface (RQ5)
(a) SCR vs.\ topology depth
Safe Completion Rate
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
0.40
2
3
5
topology depth (\# agents)
chain
star
mesh
(b) V-ID vs.\ V-IC trade-off across shapes
violation rate (depth = 3)
0.25
0.20
0.15
0.10
0.05
0.00
V-ID rate
V-IC rate
chain
star
mesh